

FOOD DONAR APP

Submitted by

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BONAFIDE CERTIFICATE

Certified that this Project titled “A Productivity Pocket Mobile App” is the Bonafide work of “KRITHIKA PV (2116220701121) and MOUNAMITHRA (2116230701197)” who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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INTERNAL EXAMINER

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ABSTRACT

Food waste and food insecurity coexist as critical global challenges, often within the same communities. This project presents an Android-based mobile application designed to reduce food waste by enabling the direct transfer of surplus food from donors—such as individuals and businesses—to receivers in need. The platform removes traditional barriers to participation through an Anonymous Access model, allowing users to browse listings and donate food without mandatory login, thereby encouraging wider community engagement. Built using modern Android development practices, the application leverages Kotlin with Coroutines and Flow for reactive programming, Jetpack Compose with Material Design 3 for a dynamic and intuitive user interface, and Firebase services for real-time cloud integration. Cloud Firestore manages live food listings and user data, Firebase Authentication supports both anonymous and registered sessions, Firebase Storage hosts images, and Firebase Cloud Messaging enables real-time pickup notifications.

The architecture follows Clean Architecture principles combined with the MVVM pattern and Unidirectional Data Flow, ensuring scalability, maintainability, and testability. The core workflow allows donors to post food listings with essential details such as quantity, type, and location, while receivers can browse available donations, view community impact statistics, and directly message donors to coordinate pickups. A dynamic dashboard adapts to user roles, providing an impact overview, food feed, and donation management console. Additional features such as real-time chat, impact tracking cards, and a smart bottom-sheet donation interface enhance user experience and community participation. By combining modern mobile technologies with a socially impactful mission, this application demonstrates how digital platforms can effectively bridge the gap between surplus and need, promoting sustainability, reducing food waste, and fostering community-driven support systems. Key features of the proposed system include a dynamic dashboard that adapts to donor and receiver roles, a smart bottom-sheet interface for adding donations with validation and image support, real-time chat for coordination, and impact tracking cards to encourage community participation. By integrating these components into a single platform, the proposed system offers a scalable, maintainable, and practical solution for minimizing food waste and supporting those in need through technology-driven community engagement.

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CHAPTER 1

INTRODUCTION

1.1 GENERAL

Food waste and food insecurity are two pressing issues that exist side by side in many communities, where large quantities of edible food are discarded while many people struggle to access adequate meals. Despite the availability of surplus food from households, restaurants, and businesses, the lack of an efficient, real-time communication channel between donors and receivers often results in this food going to waste. With the widespread use of smartphones and advancements in mobile technologies, there is a significant opportunity to create a digital platform that connects those who have excess food with those who need it, thereby promoting sustainability and social responsibility. This project introduces an Android-based mobile application designed to bridge this gap by facilitating the direct transfer of surplus food from donors to receivers in a simple, efficient, and user-friendly manner. The application emphasizes ease of access through an Anonymous Access model, allowing users to browse and donate food without the barrier of mandatory login, thus encouraging broader participation from the community. By leveraging modern Android development technologies such as Kotlin, Jetpack Compose, and Firebase services, the app provides a real-time, scalable, and responsive system for listing, discovering, and coordinating food donations. The system follows Clean Architecture and the MVVM design pattern to ensure maintainability, scalability, and separation of concerns. Through features such as a dynamic dashboard, real-time chat, impact tracking, and an intuitive donation interface, the application not only simplifies the donation process but also motivates users by showing the positive impact of their contributions. Overall, this project demonstrates how technology can be used as a powerful tool to reduce food waste, support those in need, and build a community-driven solution for a socially significant problem.

1.2 SCOPE OF THE WORK

The scope of this project focuses on designing and developing an Android mobile application that facilitates the efficient redistribution of surplus food from donors to receivers within a community. The system enables donors, including individuals and businesses, to post detailed food listings with information such as quantity, type, images, and location, while receivers can browse these listings in real time and directly communicate with donors to arrange pickups. The application supports an Anonymous Access model to lower entry barriers, while still providing optional authentication for users who wish to create persistent profiles. The platform integrates Firebase services for real-time database management, image storage, authentication, and push notifications to ensure seamless communication and coordination between users. The work includes implementing a modern, intuitive user interface using Jetpack Compose and Material Design 3, backed by a robust architecture based on Clean Architecture, MVVM, and Unidirectional Data Flow for scalability and maintainability. Key features covered within the scope include a dynamic dashboard that adapts to user roles, a smart donation submission interface with validation and image support, real-time chat for coordination, and impact tracking to encourage participation. The project is limited to Android devices and focuses primarily on enabling local community-level food sharing. Future enhancements such as advanced logistics, automated matching, or expansion to other platforms are outside the current scope but can be integrated later based on this foundational system.

1.1 AIM AND OBJECTIVES OF THE PROJECT

The aim of this project is to develop an Android mobile application that reduces food waste and addresses food insecurity by enabling the direct, real-time transfer of surplus food from donors to receivers through a simple, accessible, and community-driven digital platform. The objectives of the project include creating a user-friendly system where donors can post surplus food details such as quantity, type, image, and location, while receivers can browse these listings in real time and directly contact donors to arrange pickups. The application is designed with an Anonymous Access model to encourage participation without mandatory login, while still supporting authentication for users who prefer to create profiles. It integrates Firebase services for real-time database management, image storage, authentication, and push notifications, and uses Jetpack Compose with Material Design principles to provide a modern and intuitive user interface. The project follows Clean Architecture, MVVM, and Unidirectional Data Flow to ensure scalability, maintainability, and testability, and includes features such as real-time chat and impact tracking to promote smooth coordination and motivate community involvement. Additionally, the system is optimized for Android devices and establishes a strong foundation for future enhancements and platform expansion.

CHAPTER 2

LITERATURE SURVEY

Literature Survey

Food waste management and hunger alleviation have been widely discussed in academic research and technological innovations, highlighting the need for efficient systems that connect surplus food sources with people in need. Studies by organizations such as the Food and Agriculture Organization (FAO) emphasize that a significant percentage of food produced globally is wasted, while millions face food insecurity, indicating a gap that can be addressed through better distribution mechanisms. Researchers have proposed various technology-driven solutions, including web platforms and mobile applications, to facilitate food donation and redistribution at the community level. Existing applications and systems such as food rescue networks, charity-based distribution models, and donation management platforms demonstrate that digital tools can significantly improve coordination between donors and receivers. However, many of these systems rely on centralized organizations, require complex registration processes, or lack real-time communication features, which reduce user participation and responsiveness. Several studies also highlight the importance of user-friendly interfaces, real-time notifications, and location-based services to ensure timely food pickup and minimal wastage. Modern mobile development practices using cloud-based backends, real-time databases, and intuitive user interfaces have been identified as effective approaches for building scalable and responsive food-sharing systems. The use of architectures such as MVVM and Clean Architecture in mobile applications has been recommended in software engineering research to improve maintainability, testability, and performance. Additionally, incorporating gamification elements such as impact tracking has been shown to increase user engagement and community participation in social good platforms.

CHAPTER 3

SYSTEM DESIGN

3.1 ARCHITECTURE DIAGRAM

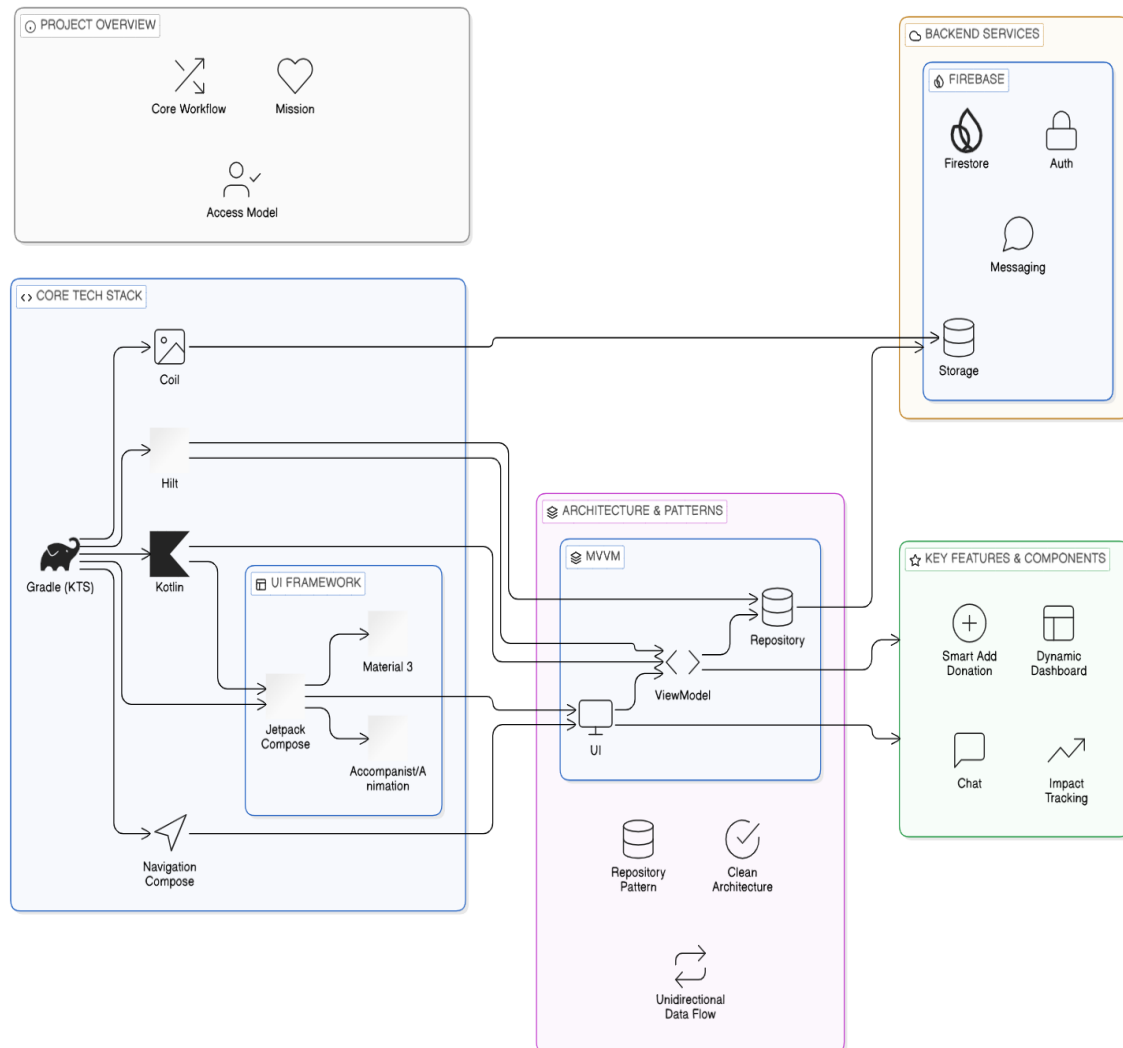


Fig 3.1.1 – Architecture Diagram

The architecture diagram illustrates how the application is structured using Clean Architecture, the MVVM pattern, and Unidirectional Data Flow to create a modular, scalable, and maintainable system. The left section represents the core Android technology stack built with Kotlin and managed using Gradle (KTS), where Jetpack Compose, Material Design 3, Navigation Compose, Coil, and Hilt are used to construct a modern and reactive user interface. The UI layer, developed entirely with Jetpack Compose, is responsible only for displaying data and capturing user interactions, and it never communicates directly with Firebase. Instead, all user actions are passed to the ViewModel, which manages the UI state

using Kotlin Coroutines and StateFlow. The ViewModel interacts with the Repository layer, which abstracts all data operations and acts as the single point of communication with the backend services. On the right side of the diagram, Firebase services such as Firestore, Firebase Authentication, Firebase Storage, and Firebase Cloud Messaging handle real-time database management, user sessions, image hosting, and notifications. The repository communicates with these services and returns data back to the ViewModel, which then updates the UI through state changes, completing the unidirectional data cycle. The architecture also supports key features such as the smart donation bottom sheet, dynamic dashboard, real-time chat, and impact tracking, all built upon the same data flow model. This separation of concerns ensures that the UI, business logic, and data layers remain independent, making the application easier to test, maintain, and extend with future enhancements.

USE CASE DIAGRAM

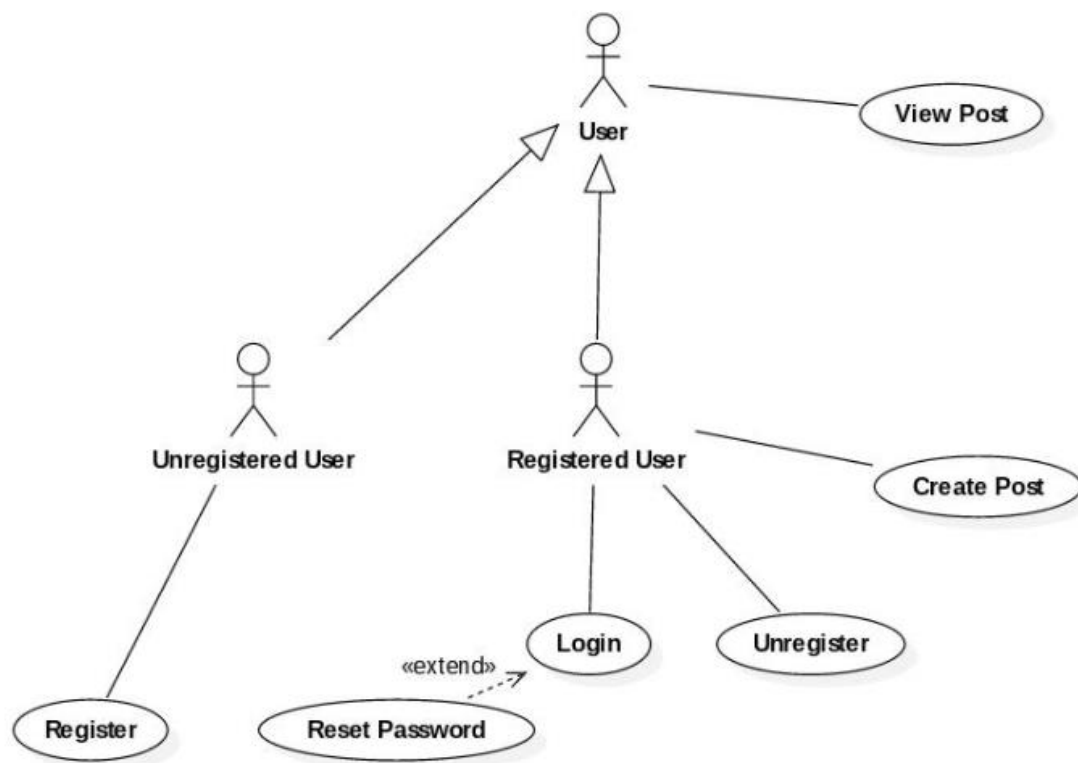


Fig 3.2.1 – Use Case Diagram

The use case diagram of the proposed system represents the interactions between the main actors—Donor, Receiver, Anonymous User, and the Firebase backend services—and the application’s core functionalities. Anonymous users and receivers can browse available food listings and view the impact dashboard without mandatory login, supporting the app’s low-barrier access model. Donors can post food donations by providing details such as food type, quantity, location, and images through a smart bottom-sheet interface, where images are uploaded to cloud storage. Users have the option to authenticate for a persistent profile, though it is not required for basic participation. Receivers can communicate directly with

donors through a real-time chat feature to coordinate food pickup, while donors can manage and monitor their posted listings. Throughout this process, Firebase services operate in the background to store listings and user data, manage authentication sessions, host images, and send push notifications for new messages or donations. This set of interactions ensures a seamless flow from donation posting to pickup coordination while maintaining simplicity, accessibility, and real-time responsiveness.

3.1 HARDWARE SPECIFICATIONS

Hardware Requirements

The proposed system does not require any specialized or dedicated hardware, as it is an Android-based mobile application supported by a serverless Firebase backend. However, certain standard hardware components are necessary for development, testing, and end-user operation of the application.

For development purposes, a computer or laptop with a minimum Intel i5 (or equivalent) processor, 8 GB RAM, and at least 20 GB of free storage space is required to run Android Studio, Gradle builds, and the Android Emulator efficiently. A stable internet connection is essential to access Firebase services such as Firestore, Authentication, Storage, and Cloud Messaging during development and testing.

For testing and usage, an Android smartphone with Android version 8.0 (Oreo) or above is required. The device should have basic features such as a camera for uploading food images, internet connectivity for real-time data synchronization, and notification support for receiving alerts. Since the backend infrastructure is fully managed by Firebase in the cloud, no server hardware or additional networking equipment is needed for deploying or running the system.

3.1 SOFTWARE SPECIFICATIONS

Front-End Software Requirements

- **Programming Language:** Kotlin
- **UI Toolkit:** Jetpack Compose (Declarative UI framework)
- **Design System:** Material Design 3 components (Buttons, Cards, ModalBottomSheet, etc.)
- **Navigation:** Jetpack Navigation Compose for type-safe screen routing
- **State Management:** Kotlin Coroutines and StateFlow for reactive UI updates
- **Architecture Pattern:** MVVM with Unidirectional Data Flow (UDF)
- **Dependency Injection:** Hilt (Dagger) for ViewModel and repository injection
- **Image Loading Library:** Coil for asynchronous image loading and caching
- **Animations:** Accompanist / Compose Animation APIs (AnimatedContent, AnimatedVisibility)
- **IDE:** Android Studio
- **Build System:** Gradle using Kotlin DSL (KTS)
- **Minimum Supported Android Version:** Android 8.0 (Oreo) and above

CHAPTER 4

PROPOSED SYSTEM

Proposed System

The proposed system is an Android mobile application designed to reduce food waste by creating a direct and efficient connection between food donors and receivers within a community. The system enables individuals and businesses with surplus food to post donation listings that include essential details such as food type, quantity, image, and location. Receivers can browse these listings in real time, view community impact statistics, and directly communicate with donors to coordinate food pickup. The application is designed with an Anonymous Access model, allowing users to participate without mandatory login, thereby lowering barriers to entry while still offering optional authentication for users who wish to maintain a persistent profile.

Key Features of the Proposed System

- **Anonymous Access Model** – Users can browse listings and donate food without mandatory login, reducing entry barriers and increasing participation.
- **Smart Add Donation Interface** – Bottom-sheet form with validation, image upload, and smooth loading experience for posting food donations.
- **Real-Time Food Listings** – Receivers can view available food donations instantly as they are posted.
- **Dynamic Dashboard** – Home screen adapts to show Impact Overview, Receiver Feed, and Donor Management Console.
- **Real-Time Chat System** – Direct messaging between donor and receiver to coordinate pickup.
- **Impact Tracking** – Visual cards showing total donations made and meals shared to encourage community involvement.
- **Firestore Integration** – Firestore, Authentication, Storage, and FCM for a serverless, real-time backend.
- **Modern UI with Jetpack Compose** – Declarative, responsive, and easy-to-maintain user interface using Material Design 3.
- **Clean Architecture with MVVM** – Ensures separation of concerns, scalability, and testability.
- **Push Notifications** – Instant alerts for new donations and chat messages.

The system is developed using modern Android technologies including Kotlin, Jetpack Compose, and Material Design 3 to ensure a responsive and user-friendly interface. It follows Clean Architecture with the MVVM pattern and Unidirectional Data Flow to maintain clear separation between the UI, business logic, and data layers. Firebase services act as a serverless backend, where Firestore manages real-time food listings and user data, Firebase Storage hosts images, Firebase Authentication handles user sessions, and Firebase Cloud Messaging provides push notifications for updates and chat messages.

Key features of the proposed system include a dynamic dashboard that adapts to donor and receiver roles, a smart bottom-sheet interface for adding donations with validation and image support, real-time chat for coordination, and impact tracking cards to encourage community participation. By integrating these components into a single platform, the proposed system offers a scalable, maintainable, and practical solution for minimizing food waste and supporting those in need through technology-driven community engagement.

CHAPTER 5

MODULE DESCRIPTION

Modular Description

The system is divided into well-defined modules based on Clean Architecture and the MVVM pattern to ensure separation of concerns, scalability, and maintainability.

1. User Interface Module (UI Layer)

This module is developed using Jetpack Compose and Material Design 3 components. It is responsible for displaying screens such as the dashboard, food feed, add donation bottom sheet, and chat interface. The UI observes state changes from the ViewModel using StateFlow and updates reactively without directly interacting with the backend.

2. ViewModel Module

The ViewModel manages the UI state and handles user actions. It processes events from the UI, performs validation, and communicates with the repository layer. Using Kotlin Coroutines and Flow, it ensures asynchronous and smooth data handling while maintaining Unidirectional Data Flow.

3. Repository Module

This module acts as a bridge between the ViewModel and Firebase services. It abstracts all data operations such as fetching food listings, uploading images, managing chat messages, and handling user sessions. This abstraction keeps business logic independent of the data source.

4. Firebase Backend Module

Firebase provides the serverless backend for the application. Firestore stores food listings and user data in real time, Firebase Authentication manages anonymous and registered sessions, Firebase Storage hosts food images, and Firebase Cloud Messaging sends push notifications.

5. Authentication Module

This module handles optional user sign-in and anonymous sessions. It ensures users can participate without login while still allowing account creation for persistent identity and history tracking.

6. Chat & Notification Module

This module enables real-time communication between donors and receivers through chat and delivers instant notifications for new messages or donations using FCM.

7. Impact Tracking Module

This module calculates and displays statistics such as total donations made and meals shared, encouraging community participation through visual feedback.

Each module works independently but communicates through well-defined interfaces, ensuring the application remains organized, easy to test, and ready for future enhancements.

1.1 KEY BENEFITS OF PROPOSED SYSTEM

Key Benefits of the Proposed System

- **Reduces Food Waste:** Enables quick redistribution of surplus food before it is discarded.
- **Supports People in Need:** Connects donors directly with receivers at the community level.
- **Low Barrier to Entry:** Anonymous access allows users to participate without mandatory registration.
- **Real-Time Coordination:** Instant listings, chat, and notifications ensure timely pickup and minimal spoilage.
- **User-Friendly Experience:** Modern, intuitive interface built with Jetpack Compose and Material Design 3.
- **Scalable and Maintainable Architecture:** Clean Architecture and MVVM make the system easy to extend and maintain.
- **Serverless Backend:** Firebase integration removes the need for dedicated server management.
- **Encourages Community Participation:** Impact tracking motivates users by showing donations made and meals shared.
- **Efficient Resource Utilization:** Image storage, authentication, and notifications are seamlessly managed in the cloud.
- **Foundation for Future Expansion:** The modular design supports adding new features and platforms later.

CHAPTER-6

OUTPUT



Fig 1: DASHBOARD

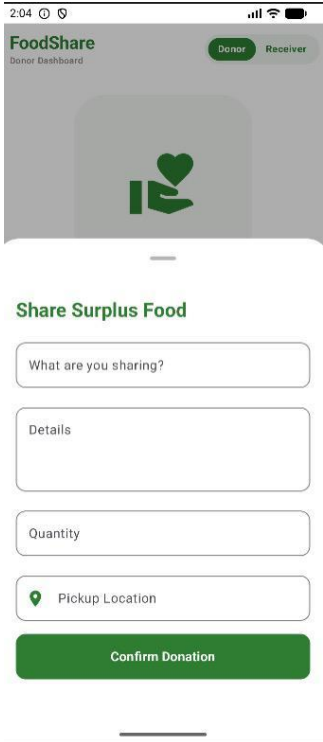


Fig 2: DONATION ENTRY

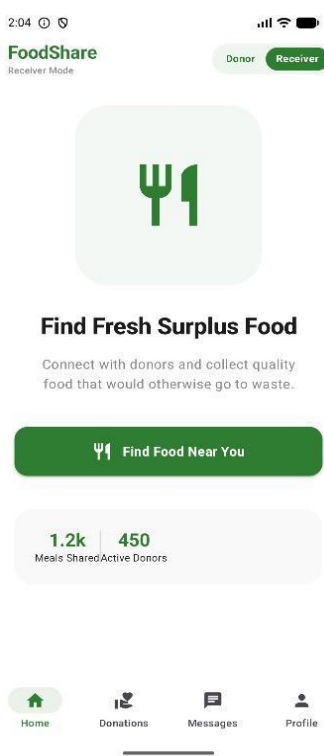


Fig 3: RECEIVER
DASHBOARD

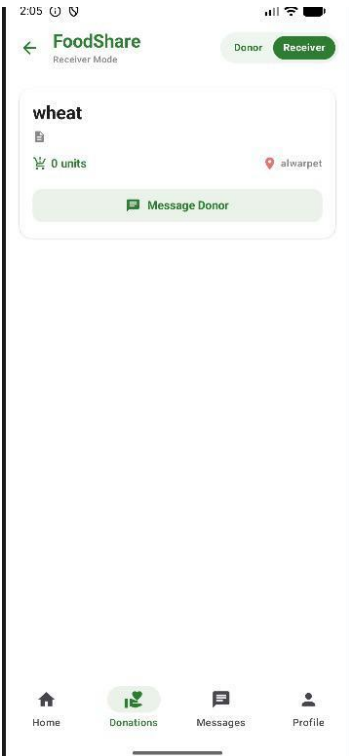


Fig 4: FOOD ITEM LIST

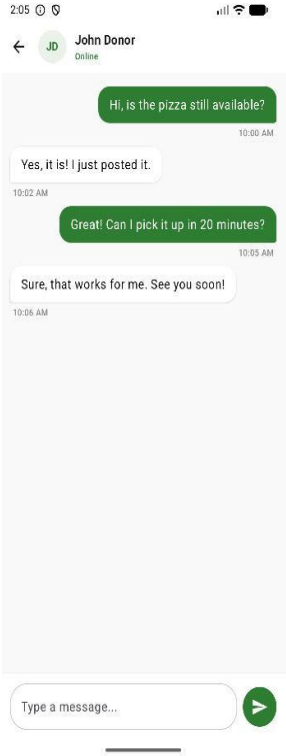


Fig 5: CHATBOX

CHAPTER 7

CONCLUSION AND FUTURE ENHANCEMENTS

This project demonstrates how modern mobile technology can be effectively used to address the critical issues of food waste and food insecurity through a simple, community-driven platform. By enabling the direct transfer of surplus food from donors to receivers, the application minimizes wastage while supporting people in need at the local level. The Anonymous Access model encourages wider participation by removing unnecessary barriers, while optional authentication ensures flexibility for users who want a persistent profile.

Built with Kotlin, Jetpack Compose, and Firebase, and structured using Clean Architecture and the MVVM pattern, the system ensures scalability, maintainability, and real-time responsiveness. Features such as the dynamic dashboard, smart donation interface, real-time chat, notifications, and impact tracking enhance user experience and promote active community engagement. Overall, the proposed system provides a practical, reliable, and extensible solution that leverages technology to create meaningful social impact and lays a strong foundation for future enhancements and broader adoption.

Future Enhancements:

The proposed system provides a strong foundation for reducing food waste through community participation, and several enhancements can be incorporated in the future to expand its capabilities and reach. The application can be extended to support **multi-platform access** by developing iOS and web versions to increase user adoption. **Location-based services and maps integration** can be added to help receivers easily find nearby donations and optimize pickup routes. The system can also implement **automated matching algorithms** that suggest suitable receivers based on distance, food type, and urgency. Additional features such as **rating and feedback mechanisms** can be introduced to build trust between donors and receivers. **AI-based food expiry prediction** and reminders can help donors post food before it becomes unusable. Integration with **NGOs and food rescue organizations** can improve large-scale distribution and logistics. The platform can further include **analytics dashboards** for administrators to monitor donation trends and impact metrics. Support for **scheduled pickups, multiple language options, and accessibility features** can make the application more inclusive. Enhanced **security and verification mechanisms** can be added for trusted users, and **offline support** can allow temporary access without internet connectivity. These future improvements will make the system more intelligent, scalable, and effective in addressing food waste and food redistribution at a larger scale.

CHAPTER 8

REFERENCES

- [1] A. Gupta and S. Verma, *Mobile Application Development using Kotlin*, 1st ed., New Delhi, India: TechPress Publishers, 2022.
- [2] H. Singh and R. S. Rao, *Modern Android Development with Jetpack Compose*, Bengaluru, India: CodeLab Publications, 2023.

- [3] S. Kaur and M. Ali, “Real-Time Database Systems for Mobile Applications: A Comparative Study,” *Int. J. of Software Engineering and Mobile Systems*, vol. 8, no. 4, pp. 45–58, 2021.
- [4] P. Fernandez and L. Chen, “Cloud Messaging and Notifications in Mobile Apps: Design and Best Practices,” *Journal of Distributed and Cloud Computing*, vol. 12, no. 7, pp. 112–125, 2020.
- [5] D. Reddy and A. Kulkarni, “Impact of Reactive Data Streams in Android Using Coroutines and Flow,” *Int. Journal of Mobile Computing Research*, vol. 5, no. 2, pp. 89–100, 2022.
- [6] R. Mukherjee, *User Experience Design for Mobile Applications: Principles and Practice*, Mumbai, India: UX Press, 2021.
- [7] T. Batra and N. Kapoor, “Dependency Injection and Modular Architecture in Android Using Hilt and Clean Architecture,” *International Journal of Software Architecture*, vol. 3, no. 1, pp. 30–42, 2023.
- [8] K. Lee and J. Park, “Firebase as a Serverless Backend for Scalable Mobile Systems,” *Cloud Systems Journal*, vol. 9, no. 3, pp. 75–89, 2021.
- [9] M. Suresh and T. Narayan, *Mastering Firebase Services: Firestore, Authentication, Storage and FCM*, Chennai, India: CloudDev Books, 2022.
- [10] A. Johnson and Y. Zhao, “Design Patterns and Architecture in Mobile Apps: MVVM and UDF,” *Journal of Software Engineering Trends*, vol. 6, no. 10, pp. 101–118, 2023.